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Bradycardia in children

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INTRODUCTION

Bradycardia is defined as a heart rate below the normal range for age (table 1). Bradycardia can be caused by intrinsic mechanisms (ie, dysfunction or injury to the heart's conduction system) or extrinsic mechanisms (ie, external factors acting on a normal heart and its conduction system) (table 2)

Children presenting with bradycardia and poor perfusion require immediate intervention (algorithm 1), which takes precedence over any diagnostic evaluation. In stable patients, the evaluation is aimed at identifying the type of rhythm disturbance and its underlying cause.

The causes, evaluation, and management of bradycardia in children will be discussed here. Other types of irregular heart rate in children and principles of pediatric advanced life support are discussed separately. (See "Irregular heart rhythm (arrhythmias) in children" and "Pediatric advanced life support (PALS)".)

DEFINITION

The normal range for heart rate varies with age (table 1) [1]. Younger patients have higher heart rates that decrease to adult values by late adolescence.

Various thresholds are used to define bradycardia in children. Within this topic review, we define bradycardia as a heart rate measured in the awake state that is below the normal range for age:

- **Infants <12 months** – Heart rate <100 beats per minute (bpm)

- **Toddlers and young children** – Heart rate <80 bpm
- **School age children** – Heart rate <70 bpm
- **Adolescents** – Heart rate <60 bpm

Alternate definitions are used in certain circumstances. For example, criteria used in automated interpretation of pediatric 12-lead electrocardiograms are generally based on the lower limit of normal (2nd percentile) for age [2-4]:

- 0 to 3 months – <126 bpm
- 3 to 6 months – <116 bpm
- 6 to 12 months – <106 bpm
- 1 to 3 years – <97 bpm
- 3 to 5 years – <77 bpm
- 5 to 8 years – <64 bpm
- 8 to 12 years – <59 bpm
- 12 to 16 years – <53 bpm

Criteria used in 24-hour ambulatory monitoring include slower heart rates that occur normally during sleep [5-8]:

- Newborns to 2 years – <60 bpm while asleep and <80 bpm while awake
- 2 to 6 years – <60 bpm
- 6 to 11 years – <45 bpm
- Adolescents >11 years – <40 bpm
- Adolescents >11 years who are well-trained athletes – <30 bpm

CLINICAL PRESENTATION

The clinical presentation of bradycardia in children can range from an incidental finding in an asymptomatic patient to severe life-threatening hemodynamic compromise.

Unstable patients — In patients with severe bradycardia, cardiac output may be insufficient, leading to poor systemic perfusion, shock, and, ultimately, cardiorespiratory arrest. Children presenting with bradycardia and poor perfusion or hypotension require immediate cardiopulmonary resuscitation (algorithm 1). (See 'Patients with hemodynamic compromise' below.)

Severe bradycardia can be due to both sinus and atrioventricular (AV) node dysfunction. This most commonly results from hypoxia, hypotension, and metabolic acidosis; however, intrinsic conduction abnormalities (eg, complete heart block) can also cause severe bradycardia [9]. (See 'Causes of sinus bradycardia' below and 'Atrioventricular heart block' below.)

Stable patients — Patients with milder degrees of bradycardia are usually asymptomatic [5,10]. Symptoms, when present, differ depending on age and whether there is underlying cardiac disease:

- **Infants and young children** – Dizziness and syncope are difficult to ascertain in infants and preverbal young children. Thus, symptomatic patients at this age often present with nonspecific symptoms of poor feeding and lethargy. Syncope and seizure-like episodes may occur if cerebral perfusion is abruptly decreased due to a slow heart rate [11,12].
- **Children and adolescents** – Among children and adolescents, bradycardia can present with fatigue, exercise intolerance, dizziness, and/or syncope [10,13].
- **Underlying congenital heart disease** – In children with underlying congenital cardiac disease, bradycardia most commonly does not produce any symptoms [5,10]. However, this group is more susceptible to having symptomatic bradycardia as they may be less able to tolerate a low heart rate due to underlying compromised cardiac function.

ACUTE ASSESSMENT AND STABILIZATION

Assessment of the patient's hemodynamic status takes precedence over any diagnostic evaluation. (See "Assessment of systemic perfusion in children".)

Patients with hemodynamic compromise — Children presenting with bradycardia associated with hemodynamic compromise (poor perfusion, shock, hypotension) require immediate intervention.

Management should proceed according to the following steps (algorithm 1) [14,15]:

- Assess the airway and circulatory system. Provide airway management, oxygenation, and ventilation as needed. (See "Pediatric advanced life support (PALS)", section on 'Assessment' and "Basic airway management in children" and "Pediatric basic life support (BLS) for health care providers", section on 'Breathing'.)
- Begin chest compressions if the heart rate remains <60 beats per minute (bpm) and there is ongoing hemodynamic compromise despite initial airway/ventilation interventions. (See "Pediatric basic life support (BLS) for health care providers", section on 'Chest compressions'.)

Pediatric bradycardia with a pulse 2020 update

Algorithm 1 - larger image below

- Administer a medication to increase the heart rate:
  - Epinephrine for patients without increased vagal tone or primary atrioventricular (AV) block – Administer epinephrine at a dose of 0.01 mg/kg intravenously (IV) or intraosseously (IO). The dose may be repeated every three to five minutes at the same dose. The maximum single dose is 1 mg. (See "Primary drugs in pediatric resuscitation", section on 'Epinephrine'.)

    If IV or IO access is not available and the patient is intubated, epinephrine can be administered through the endotracheal tube using a higher dose of 0.1 mg/kg.
  - Atropine for patients with increased vagal tone or primary AV block – Administer atropine at a dose of 0.02 mg/kg IV or IO (minimum dose 0.1 mg, maximum dose 0.5 mg). The dose may be repeated once. (See "Primary drugs in pediatric resuscitation", section on 'Atropine'.)
- Identify and treat potential reversible causes of bradycardia, including hypoxia, hypothermia, hypovolemia, head injury, medications/toxins, and hypervagotonia. (See 'Causes of sinus bradycardia' below.)
- If bradycardia persists, consider transthoracic or transvenous cardiac pacing, particularly if a conduction defect is suspected or confirmed (eg, complete heart block). Cardiac pacing requires that appropriately trained personnel and equipment are available. Temporary pacing is not helpful for patients with refractory bradycardia due to ongoing hypoxia or ischemia.

This stepwise approach is consistent with the American Heart Association (AHA) guidelines for pediatric advanced life support (PALS) [14]. Additional principles of PALS are discussed separately. (See "Pediatric advanced life support (PALS)".)

In a registry study of nearly 2800 hospitalized pediatric patients who received CPR for bradycardia with poor perfusion, 31 percent became pulseless after CPR was started (median time to loss of pulses was three minutes) [16]. Children who became pulseless were more likely to have metabolic abnormalities, septicemia, malignancy, or other organ system dysfunction (hypotension, heart failure, kidney function impairment) at the time of the arrest. The duration of CPR was considerably longer for children who became pulseless during the resuscitation (20 versus 4 minutes) and the likelihood of surviving to hospital discharge was lower (30 versus 70 percent). These findings highlight the importance of prompt recognition and treatment of underlying reversible causes of bradycardia with poor perfusion, as well as prompt initiation of good quality CPR followed by appropriate post-resuscitation care. (See "Pediatric advanced life support (PALS)", section on 'Early post-cardiac arrest management' and "Initial post-cardiac arrest care in children", section on 'Approach to stabilization'.)

Stable patients — For patients who are hemodynamically stable, additional time can be given to determine the type of rhythm abnormality and identify the underlying cause, as discussed in the following sections.

APPROACH TO EVALUATION IN STABLE PATIENTS

Clinical evaluation — In children with bradycardia who are hemodynamically stable, the evaluation is aimed at identifying the type of rhythm disturbance and its underlying cause. The evaluation begins with a history, physical examination, and electrocardiogram (ECG).

History — The history should include the following elements, which are aimed at identifying possible causes of bradycardia (table 2):

- History of exposure to medications that cause bradycardia (table 3), including prescribed medications as well as possible accidental ingestions
- Family history of syncope or sudden cardiac death
- History of cardiac disease
- Past episodes of syncope, dizziness, and/or unexplained seizures

Physical examination — Important elements of the physical examination include:

- **Heart rhythm** – An irregular heartbeat that arises from a pause in an otherwise regular rhythm suggests the possibility of sinus node dysfunction or second-degree atrioventricular (AV) block

- **Abnormal heart sounds** – The presence of a heart murmur or gallop may suggest underlying congenital heart disease or other cardiac pathology

Cardiac tests — The following tests are useful in the evaluation of patients with bradycardia:

- **ECG** – A 12-lead ECG should be obtained in children found to have bradycardia on physical examination and in those with concerning symptoms (ie, syncope, dizziness, and/or unexplained seizures). The 12-lead ECG will often help determine the type of rhythm abnormality, although its use may be limited by the occurrence of bradycardia as a transient event. (See 'Identifying the type and cause of bradycardia' below.)

- **Ambulatory ECG monitoring** – While ambulatory ECG monitoring is not routinely necessary for all children presenting with bradycardia, it may be useful in select circumstances. Examples include:

  - Patients with second- or third-degree AV block detected by 12-lead ECG.

- Patients presenting with syncope/near-syncope or other concerning symptoms, in cases where bradycardia, sinus node dysfunction, or heart block is suspected as the etiology. In this setting, ambulatory ECG monitoring can assess physiologic heart rate response.

- Patients with genetic conditions predisposing to bradycardia (eg, long QT syndrome, cardiomyopathy). (See "Congenital long QT syndrome: Diagnosis".)

- Patients with certain congenital heart lesions and/or postoperative conditions predisposing to bradycardia (eg, Fontan procedure). (See "Atrial arrhythmias and atrioventricular block associated with congenital heart disease", section on 'Bradycardias'.)

- Patients recently initiated heart rate slowing medication.

The Holter monitor is a commonly used device for ambulatory ECG monitoring in children. It provides a continuous rhythm recording from adhesive electrodes for a minimum of 24 or 48 hours. It uses a small, lightweight, battery-operated electromagnetic tape recorder or digital recorder that records two or three channels of ECG data, including minimum, average, and maximum heart rate and longest pause in heart rate.

Other ambulatory ECG monitors are available and may be useful for patients with paroxysmal symptoms. These monitors include cardiac event recorders, which can be either looping (historical ECG data) or nonlooping.

For longer-term monitoring, there are wearable patches that can provide data for up to 14 days [17]. Insertable cardiac monitors may also be used for long-term monitoring in select patients [18].

Technologic advancements have expanded the types of wearable monitors on the market and include devices such as smartphone wireless ECG monitors and wrist-worn wearables, such as the Apple Watch [19,20].

Additional details regarding ambulatory ECG monitoring are provided separately. (See "Ambulatory ECG monitoring".)

- **Echocardiogram** – Although rarely required in the evaluation of children with asymptomatic sinus bradycardia, transthoracic echocardiography may be useful in certain patients to assess ventricular function and identify structural abnormalities.

Indications for echocardiography in children with bradycardia may include [21]:

- Syncope with abnormal ECG findings

- Syncope in a patient with family history of sudden cardiac death or cardiomyopathy
- Exertional syncope
- Follow-up evaluation of children with chronic bradycardia to assess for progressive ventricular enlargement

- **Exercise stress testing** – Exercise stress testing may be used to determine heart rate response with activity (chronotropic competence). It is helpful in differentiating abnormal cardiac conduction from hypervagotonia, as the former fails to respond to exercise, while the latter is able to achieve normal peak heart rates. (See "Exercise ECG testing: Performing the test and interpreting the ECG results".)

  Chronotropic incompetence (the inability to appropriately respond to stress) is a feature of sinus node dysfunction that may manifest as dyspnea or lightheadedness on exertion. (See "Sinus node dysfunction: Clinical manifestations and diagnosis", section on 'Patients with exertional symptoms'.)

- **Invasive electrophysiology testing** – An invasive electrophysiology study (EPS) is rarely required as part of the evaluation. An EPS may be helpful if the mechanism for the arrhythmia remains unclear or if symptoms suggest the presence of a potentially life-threatening arrhythmia [22]. (See "Invasive diagnostic cardiac electrophysiology studies".)

IDENTIFYING THE TYPE AND CAUSE OF BRADYCARDIA

The type of rhythm disturbance (sinus bradycardia versus atrioventricular [AV] block) can usually be determined based upon the electrocardiogram (ECG) findings. Identifying the type of rhythm abnormality helps to narrow the list of likely causes.

Pathophysiology

- **Control of heart rate** – Heart rate is controlled by both the cardiac conduction system and the nervous system:

  - **Conduction system** – The role of the cardiac conduction system is to initiate and conduct the electrical signal that controls and coordinates atrial and ventricular contraction, as reflected in the electrocardiogram (ECG) (figure 1) [23]. The components of the conduction system include the sinus node, AV node, and His-Purkinje system (His bundle). (See "ECG tutorial: Physiology of the conduction system".)

    The sinus node is the pacemaker of the heart and is located in the sulcus terminalis at the junction of the superior vena cava and the right atrium. Specialized cells (nodal or dominant cells) within the sinus node depolarize spontaneously, which initiates an electrical impulse that spreads to the AV node and through the atria, resulting in bilateral atrial contraction [5]. At the AV node, the impulse is conducted through to the bundle of

His. The His bundle courses through the region of the membranous septum and divides thereafter into the right and left bundle branches. Electrical activation through an intact His bundle and bundle branches results in nearly simultaneous contraction of the right and left ventricles. (See "Normal sinus rhythm and sinus arrhythmia", section on 'Anatomy'.)

- **Nervous system** – Both the parasympathetic and sympathetic nervous system innervate the cardiac conduction system.

  Increased parasympathetic tone via the vagus nerve both decreases sinus node pacing rate and slows the AV node conduction leading to a decrease in heart rate. A very strong vagal response can transiently depress sinus node depolarization (sinus pause) or block transmission through the AV node (complete AV block). (See 'Hypervagotonia' below.)

  Increased sympathetic tone increases sinus node pacing leading to an increase in heart rate.

- **Mechanisms of bradycardia** — Bradycardia may be caused by intrinsic mechanisms (ie, dysfunction or injury to the heart's conduction system) or extrinsic mechanisms (ie, external factors acting on a normal heart and its conduction system) (table 2).

  There are two main mechanisms and sites for the development of bradycardia:

  - **Sinus bradycardia** – At the sinus node, the depolarization rate is decreased below the lowest normal heart rate values set for age (see 'Sinus bradycardia' below)

  - **AV node block** – Conduction of the electrical impulse is either delayed or blocked at the AV node or the bundle of His (see 'Atrioventricular heart block' below)

Sinus bradycardia — Sinus bradycardia is present when there is a normal-appearing P wave (waveform 1) and normal PR interval, but the rate is below the normal range for age (table 1).

Sinus bradycardia may be accompanied by a sinus pause (an absent P wave on the ECG tracing). Sinus pauses or absent P waves may be due either to failure to generate a sinus node depolarization or failure of a generated sinus node depolarization to exit the sinus node and enter the atria.

Sinus node dysfunction (waveform 2) represents a major subcategory of sinus bradycardia. It is characterized by inappropriately low heart rate or

Single lead electrocardiogram (ECG) showing sinus bradycardia

Waveform 1 - larger image below

chronotropic incompetence (failure to appropriately elevate the heart rate in response to physiologic stress). In addition to sinus bradycardia, other findings associated with sinus node dysfunction include severe sinus pauses, tachycardia/bradycardia syndrome, sinus node reentry tachycardia, and sinoatrial exit block. (See "Sinus node dysfunction: Epidemiology, etiology, and natural history" and "Sinus node dysfunction: Clinical manifestations and diagnosis".)

Causes of sinus bradycardia — Common causes of sinus bradycardia in children and adolescents include hypervagotonia, severe physiologic disturbances (eg, hypoxia, hypotension, hypothermia), medications, athletic training, and congenital heart disease (CHD) (table 2). In addition, sinus bradycardia is commonly observed in asymptomatic children without any apparent underlying pathology; this tends to have a benign course, as discussed below. (See 'Asymptomatic bradycardia in otherwise healthy children' below.)

Causes of bradycardia in children

Table 2 - larger image below

Hypervagotonia — Hypervagotonia (exaggerated vagal activity) slows the pacing rate of the sinus node and slows the conduction time through the atrioventricular (AV) node. Bradycardia induced by exaggerated vagal activity is usually transient and triggered by a precipitating event. Examples of common triggers include:

- Nasopharyngeal or esophageal stimulation (eg, placement of nasogastric or endotracheal tubes).
- Breath-holding spells [24,25] – In an older study of 58 children with pallid breath-holding spells, ocular compression triggered sinus pause of greater than two seconds in almost 80 percent of patients and greater than four seconds in 55 percent of patients. However, applying ocular pressure as a diagnostic test is no longer recommended, due to the risk of injury to the eye [25]. (See "Breath-holding spells".)
- Gastroesophageal reflux (GER) and vomiting – GER is commonly cited as an etiology for bradycardia in premature neonates. However, a study in preterm infants (average gestational age 29.4 weeks) showed that bradycardia events were preceded by GER in only 2.9 percent of cases [26]. (See "Gastroesophageal reflux in premature infants" and "Gastroesophageal reflux in infants".)
- Coughing.

- Obstructive sleep apnea – Studies of young adults with obstructive sleep apnea have demonstrated severe bradycardia (heart rates below 30 beats per minute [bpm]) and asystole of 10 seconds due to enhanced vagal tone. (See "Obstructive sleep apnea and cardiovascular disease in adults".)

In addition, vagally mediated bradycardia may occur due to increased intracranial pressure [27] (see "Elevated intracranial pressure (ICP) in children: Clinical manifestations and diagnosis")

In hypervagotonia-induced bradycardia, simultaneous AV nodal dysfunction may also occur and contribute to symptoms. The combination of the slow heart rate and an associated decline in peripheral vascular resistance are often sufficient to produce presyncope or syncope. The evaluation and management of individuals with neurally mediated (reflex) syncope is discussed separately. (See "Reflex syncope in adults and adolescents: Clinical presentation and diagnostic evaluation".)

Drugs and toxins — Many medications can cause bradycardia (table 3), either by direct effects on the sinus or AV node (eg, beta-adrenergic blockers, calcium channel blockers) or indirect effects through the nervous system (eg, opioids). In some cases, bradycardia can be caused by both direct and indirect effects (eg, clonidine). In neonates, bradycardia can occur due to maternal exposure to medications (eg, beta blockers) in late pregnancy [28].

Medications that can cause bradycardia in children

Table 3 - larger image below

Drugs and toxins that can cause sinus node disfunction are discussed separately. (See "Sinus node dysfunction: Epidemiology, etiology, and natural history", section on 'Medications and toxins'.)

Physiologic disturbances (hypoxia, hypotension, hypothermia) — Bradycardia can occur as a consequence of severe physiologic disturbances, such as hypoxia, hypotension, acidosis, and hypothermia. Bradycardia occurs at moderate to severe levels of hypothermia (core temperature <32°C [90°F]). (See "Hypothermia in children: Clinical manifestations and diagnosis", section on 'Pathophysiology'.)

Endurance training-related bradycardia — Well-trained endurance athletes often have resting sinus bradycardia. (See "Evaluating the athlete's electrocardiogram", section on 'Normal (ie, training-related) findings'.)

Congenital heart disease — Sinus node dysfunction can occur in various CHD lesions, most commonly atrial septal defects (ASDs). Concomitant AV nodal dysfunction is also observed in patients with ASDs [10,29,30]. (See "Atrial arrhythmias and atrioventricular block associated with congenital heart disease" and "Isolated atrial septal defects (ASDs) in children: Classification, clinical features, and diagnosis".)

Other CHD lesions associated with sinus node dysfunction include AV canal defect, pulmonary stenosis, ventricular septal defect, single ventricle with heterotaxy (particularly the "polysplenia" type), and transposition of the great arteries [10,31].

Sinus node dysfunction and associated atrial arrhythmias can also occur after surgical or transcatheter repair of CHD. This occurs most commonly in repairs to the atrial wall that injure the sinus node, sinus node artery, or innervation of the sinus node [10,30,32-34]. (See "Atrial arrhythmias and atrioventricular block associated with congenital heart disease", section on 'Acquired sinus node dysfunction'.)

Historically, atrial arrhythmias were commonly observed following atrial switch operations (Mustard or Senning procedures) for transposition of the great arteries; however, these surgeries are no longer performed in the contemporary era. (See "D-transposition of the great arteries (D-TGA): Management and outcome", section on 'Complications after Mustard and Senning procedures'.)

Other causes

- **Cardiomyopathy and myocarditis** – Cardiomyopathy and acquired inflammatory heart diseases (myocarditis, pericarditis, Kawasaki disease) can cause sinus node dysfunction. (See "Familial dilated cardiomyopathy: Prevalence, diagnosis and treatment" and "Clinical manifestations and diagnosis of myocarditis in children" and "Cardiovascular sequelae of Kawasaki disease: Clinical features and evaluation".)

- **Familial sinus node dysfunction** – Familial sinus node dysfunction is rare. Mutations in two *HCN4* or *SCN5A* are responsible for some cases [35]. (See "Sinus node dysfunction: Epidemiology, etiology, and natural history", section on 'Childhood and familial disease'.)

- **Long QT syndrome** – Sinus bradycardia os a common finding in patients with congenital long QT syndrome. (See "Congenital long QT syndrome: Epidemiology and clinical manifestations", section on 'Bradycardia'.)

- **Cardiac tumors** – Cardiac tumors are a rare cause of sinus bradycardia or sinus node dysfunction. (See "Cardiac tumors".)

- **Eating disorders** – Bradycardia is a feature seen in individuals with severe restrictive eating disorders. (See "Eating disorders: Overview of epidemiology, clinical features,

and diagnosis".)

- **Prematurity** – Bradycardic episodes associated with apnea of prematurity are thought to be due to a combined mechanism of increased vagal tone and immature sinus node function. (See "Pathogenesis, clinical manifestations, and diagnosis of apnea of prematurity".)

Atrioventricular heart block — AV block is defined as a delay or interruption in the transmission of an atrial impulse to the ventricles due to an anatomical or functional impairment in the conduction system. The conduction can be delayed, intermittent, or absent. Heart block is categorized into first-degree (slowed conduction without missed beats), second-degree (intermittent conduction with missed beats), and third-degree or complete AV block (absent conduction).

AV block can be caused by underlying CHD, acquired heart disease, medications, or it may be a complication of surgical or transcatheter intervention (table 4).

Major causes of atrioventricular (AV) block

Table 4 - larger image below

First-degree atrioventricular block — In first-degree AV block, the PR interval is greater than the upper limits of normal for age (waveform 3). The PR interval is both age- and rate-dependent. In general, the normal PR interval is 70 to 170 milliseconds in newborns and 80 to 220 milliseconds in young children and adults [36].

First-degree AV block in isolation does **not** cause bradycardia. This is because AV conduction remains intact despite the prolonged PR interval. Nevertheless, it is important to distinguish this finding from other forms of AV nodal dysfunction that do cause bradycardia.

First-degree AV block is a common ECG finding. In one study, first-degree AV block was documented in 6 percent of healthy newborns [37].

First-degree AV block occurs in association with hypervagotonia, sinus node dysfunction, or AV node disease. (See 'Hypervagotonia' above and "Sinus node dysfunction: Clinical manifestations and diagnosis".)

Medical conditions associated with first-degree AV block include [38]:

- Rheumatic fever
- Lyme disease

- Chagas disease
- Rubella, mumps
- Hypothermia
- Metabolic derangements (hypokalemia, hyperkalemia, hypocalcemia, hypercalcemia, hypoglycemia, and hypomagnesemia)
- Cardiomyopathy

Children with first-degree AV block are usually asymptomatic. Routine monitoring and/or treatment may be indicated if there is evidence of coexisting sinus or AV node dysfunction.

With extreme PR interval prolongation, some patients may develop pacemaker syndrome (dizziness, fatigue, or chest discomfort) due to atrial contraction against a closed mitral valve or when atrial contraction occurs shortly after ventricular systole with incomplete atrial filling.

Additional aspects of clinical manifestations and implications of first-degree AV block at different sites in the conduction system are discussed separately. (See "First-degree atrioventricular block".)

Second-degree atrioventricular block — In second-degree AV block, the organized atrial impulse fails to be conducted to the ventricle in a 1:1 ratio. Second-degree AV block is further divided into two categories:

Mobitz type I block — In Mobitz type I block (also referred to as Wenckebach block), there is progressive prolongation of the PR interval until a P wave fails to be conducted (waveform 4).

Mobitz type I block is commonly seen in healthy children and young adults, especially at times of high parasympathetic tone (eg, sleep or in well-trained athletes) [39,40].

Mobitz type I block rarely causes any symptoms. The block is located at the level of the AV node and is usually not associated with other significant conduction system disease. In addition, it does not progress to complete block.

It may also be seen in patients with intrinsic AV nodal disease, myocarditis (including Chagas disease), Lyme disease, or myocardial infarction or following cardiac surgery. These conditions may be associated with other conduction abnormalities.

Intervention is usually not indicated unless there is evidence for other more significant conduction system disease. (See "Second-degree atrioventricular block: Mobitz type I (Wenckebach block)".)

Mobitz type II block — In Mobitz type II block, the PR interval remains unchanged prior to a P wave that fails to conduct to the ventricles (waveform 5).(See "Second-degree

atrioventricular block: Mobitz type II".)

Mobitz type II block is far less common than type I and but its presence has more significant clinical implications [40]. It is associated with various forms of CHD and can be seen after cardiac surgery. Type II block is thought to occur at or below the level of the AV node, indicating disease within the His bundle and bundle branches. It has a less predictable course and may progress to complete heart block.

Advanced second-degree atrioventricular block — In advanced (or high-grade) second-degree AV block, two or more consecutive P waves are nonconducted to the ventricles, though some P waves are conducted. This indicates significant conduction disease below the level of the AV node.

Third-degree atrioventricular block — In third-degree AV block, also referred to as complete heart block, there is complete dissociation of the atrial and ventricular activity. The atrial rate (P wave) is greater than the ventricular rate (QRS complex), which is junctional or ventricular in origin (waveform 6). The QRS complex can be narrow (consistent with a junctional escape rhythm) or wide (consistent with a ventricular escape rhythm). (See "Acquired third-degree (complete) atrioventricular block", section on 'Diagnosis of third-degree AV block'.)

Complete heart block can be congenital or acquired.

Congenital complete heart block — Neonatal lupus, due to maternal antibodies (RO-SSA and LA-SSB) that cross the placenta, is responsible for 60 to 90 percent of congenital complete heart block cases. Other causes include myocarditis and various structural cardiac defects, particularly L-transposition of the great arteries, AV discordance, or polysplenia with AV canal defect. The clinical manifestations, presentation, and treatment of congenital complete heart block are discussed separately. (See "Congenital third-degree (complete) atrioventricular block" and "Neonatal lupus: Epidemiology, pathogenesis, clinical manifestations, and diagnosis".)

Familial AV conduction block, characterized by a progression in the degree of block in association with a variable apparent site of block, may be transmitted as an autosomal dominant trait. (See "Etiology of atrioventricular block", section on 'Familial disease'.)

Acquired complete heart block — Acquired complete heart block can be transient or permanent. Causes of acquired heart block include (table 4) (see "Acquired third-degree (complete) atrioventricular block", section on 'Etiology'):

- Myocarditis
- Lyme disease
- Acute rheumatic disease

- Myocardial infarction
- Trauma
- Injury from surgery or catheterization
- Cardiomyopathy

Patients with acquired complete heart block may present with hemodynamic instability due to the ability to maintain an adequate cardiac output. These patients are also at increased risk for sudden cardiac death. (See "Acquired third-degree (complete) atrioventricular block".)

LONG-TERM MANAGEMENT OF PATIENTS WITH CHRONIC BRADYCARDIA

Long-term management of children with chronic bradycardia depends upon the specific conduction abnormality, the frequency and severity of symptoms, and whether the child has underlying congenital heart disease (CHD). Children with sinus bradycardia who are asymptomatic and otherwise healthy generally do not require any further evaluation or treatment.

For children requiring intervention, permanent pacemaker implantation is the treatment of choice. Chronic medical therapy is generally not effective, due to variable response over time. In addition, there are unacceptable side effects of chronotropic medications. We generally agree with the recommendations of the Pediatric and Congenital Electrophysiology Society (PACES), American College of Cardiology (ACC), AHA, and Heart Rhythm Society (HRS) regarding indications for permanent pacemaker placement in pediatric patients [18]. (See 'Society guideline links' below.)

A more detailed discussion of indications for permanent pacemaker placement is provided separately. (See "Permanent cardiac pacing: Overview".)

Complete heart block — Definitive treatment of third-degree (complete) AV block generally involves permanent pacemaker placement, as discussed separately. (See "Congenital third-degree (complete) atrioventricular block", section on 'Role of pacing' and "Permanent cardiac pacing: Overview", section on 'Acquired AV block'.)

Sinus node dysfunction — The need for permanent pacing in patients with sinus node dysfunction depends upon the correlation of bradycardia with symptoms. For patients with sinus node dysfunction that correlates with symptoms during age- and activity-inappropriate bradycardia, we suggest pacemaker placement [18]. This recommendation is largely extrapolated from the experience in adult patients. There are limited pediatric data to guide management decisions [41]. (See "Permanent cardiac pacing: Overview", section on 'Sinus node dysfunction' and "Sinus node dysfunction: Treatment", section on 'Permanent pacing'.)

Congenital heart disease — Permanent pacemaker placement is indicated in patients with CHD who have acquired third-degree (complete) AV block, as is the case for patients without CHD. (See "Permanent cardiac pacing: Overview", section on 'Acquired AV block' and "Atrial arrhythmias and atrioventricular block associated with congenital heart disease", section on 'Permanent pacemaker implantation'.)

In addition, permanent pacemaker placement may be undertaken in patients with CHD in the following settings, although the evidence to support these indications is less well established [18]:

- Sinus bradycardia in patients with complex CHD who have a resting heart rate <40 bpm or prolonged pauses in ventricular rate. These patients may benefit from pacemaker placement even in the absence of associated symptoms. The rationale is based on the concern that bradycardia in these children is associated with heart failure, hemodynamic compromise, and increased risk of sudden cardiac death [42].
- Sinus bradycardia associated with recurrent episodes of intra-atrial re-entrant tachycardia when catheter ablation or medication are ineffective or not acceptable treatments
- Impaired hemodynamics due to sinus bradycardia or loss of AV synchrony
- Sinus node dysfunction, particularly with alternating tachycardia and bradycardia with symptoms attributable to pauses due to sudden-onset bradycardia

Pacemaker implantation for bradyarrhythmias is uncommon following repair of CHD. The frequency depends upon the CHD lesion and the type of repair. (See "Atrial arrhythmias and atrioventricular block associated with congenital heart disease".)

The surgeries that are most commonly associated with pacemaker placement include [32,43,44]:

- Surgical closure of atrial septal defect (up to 4 percent) (see "Isolated atrial septal defects (ASDs) in children: Management and outcome")
- Fontan (7 percent patients) (see "Management of complications in patients with Fontan circulation", section on 'Arrhythmias')
- Atrial switch operation (over 80 percent) (see "D-transposition of the great arteries (D-TGA): Management and outcome", section on 'Mustard and Senning procedures')

Asymptomatic bradycardia in otherwise healthy children — Children and adolescents with sinus bradycardia who are asymptomatic and otherwise healthy generally do not require intervention. Although data are limited, asymptomatic sinus bradycardia in healthy children and adolescents appears to have a benign course without long-term clinical sequelae. If the

child or adolescent becomes symptomatic (eg, experiences a syncopal or near syncopal episode), a thorough evaluation is warranted. Sinus node dysfunction is a possible diagnosis in this scenario. (See 'Cardiac tests' above and "Sinus node dysfunction: Clinical manifestations and diagnosis", section on 'Evaluation'.)

In a study of 104 adolescents identified with sinus bradycardia through a population-based screening program, 17 percent of the participants had self-limited syncopal episodes, but none experienced life-threatening events during the 10-year follow-up period [45]. Additional evaluation in a subgroup of patients did not detect any *HCN4* gene mutations, which are associated with sinus node dysfunction. (See "Sinus node dysfunction: Epidemiology, etiology, and natural history", section on 'Childhood and familial disease'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Arrhythmias in children" and "Society guideline links: Basic and advanced cardiac life support in children".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient education" and the keyword[s] of interest.)

- Basics topics (see "Patient education: Bradycardia (The Basics)" and "Patient education: Heart block in adults (The Basics)" and "Patient education: Heart block in children (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Definition** – Bradycardia is defined as a heart rate measured in the awake state that is below the normal range for age (table 1). (See 'Definition' above.)

- **Causes of bradycardia** – Bradycardia can be caused by intrinsic mechanisms (ie, dysfunction or injury to the heart's conduction system) or extrinsic mechanisms (ie, external factors acting on a normal heart and its conduction system). (See 'Pathophysiology' above.)

  Common causes of bradycardia in children and adolescents include hypervagotonia, physiologic disturbances (eg, hypoxia, hypotension, hypothermia), medications (table 3), athletic training, and congenital heart disease (CHD) (table 2 and table 4). In addition, sinus bradycardia is commonly observed in asymptomatic children without any apparent underlying pathology. (See 'Causes of sinus bradycardia' above and 'Atrioventricular heart block' above.)

- **Clinical presentation**

  - **Unstable patients** – If cardiac output is insufficient, the patient may present with poor systemic perfusion or cardiorespiratory arrest. (See 'Unstable patients' above.)

  - **Stable patients** – Patients with milder degrees of bradycardia most commonly are asymptomatic. In infants and preverbal children, symptoms, when present, may be nonspecific, including lethargy and poor feeding. Older children and adolescents may complain of dizziness, fatigue, exercise intolerance, and/or syncope. (See 'Stable patients' above.)

- **Acute stabilization** – Patients presenting with severe bradycardia and poor perfusion require immediate cardiopulmonary resuscitation as summarized in the algorithm (algorithm 1). Initial interventions aim to identify and address reversible causes of bradycardia (eg, hypoxia). Epinephrine and/or atropine may be required if bradycardia persists despite assisted ventilation. (See 'Patients with hemodynamic compromise' above and "Pediatric advanced life support (PALS)".)

- **Evaluation** – In the stable patient with bradycardia, the evaluation begins with a medical history, physical examination, and electrocardiogram (ECG). Additional studies that may be useful include ambulatory ECG monitoring (eg, Holter monitor or other wearable device), echocardiogram, exercise stress testing, and, rarely, invasive electrophysiology testing. (See 'Clinical evaluation' above.)

  Based upon the ECG findings, the type of rhythm abnormality can be determined (see 'Identifying the type and cause of bradycardia' above):

  - **Sinus bradycardia** – Asymptomatic sinus bradycardia (waveform 1) is a common finding in healthy children. Increased vagal tone can cause sinus bradycardia that may be symptomatic. There are numerous other causes of sinus bradycardia, as summarized in

the table (table 2) and discussed above. (See 'Sinus bradycardia' above and 'Hypervagotonia' above and 'Causes of sinus bradycardia' above.)

- **Sinus node dysfunction** – Sinus node dysfunction (waveform 2) is defined as inappropriate sinus bradycardia or inability to appropriately elevate the heart rate in response to physiologic stress. It can be caused by congenital or familial cardiac disease (eg, structural CHD, cardiomyopathy), direct injury (surgery), inflammation (myocarditis), or medications. (See "Sinus node dysfunction: Clinical manifestations and diagnosis" and "Sinus node dysfunction: Epidemiology, etiology, and natural history".)

- **First-degree (atrioventricular) AV block** – First-degree AV block (slowed conduction without missed beats (waveform 3)) is frequently observed in healthy asymptomatic children. This does **not** cause bradycardia. (See 'First-degree atrioventricular block' above.)

- **Second-degree AV block** – Second-degree AV block (intermittent conduction with missed beats) includes Mobitz types I (waveform 4) and II (waveform 5). Mobitz type I block rarely causes symptoms. Mobitz type II block is associated with underlying cardiac disease and may progress to complete heart block. (See 'Second-degree atrioventricular block' above.)

- **Complete heart block** – Third-degree or complete heart block (absent conduction (waveform 6)) can be congenital or acquired. In the neonate, the most common cause is neonatal lupus. (See 'Third-degree atrioventricular block' above.)

- **Long-term management of chronic bradycardia** – Long-term management of children with chronic bradycardia depends upon the specific conduction abnormality, the frequency and severity of symptoms, and whether the child has underlying CHD. (See 'Long-term management of patients with chronic bradycardia' above.)

  - Third-degree (complete) AV block – Definitive treatment of third-degree (complete) AV block generally involves permanent pacemaker placement, as discussed separately. (See "Congenital third-degree (complete) atrioventricular block", section on 'Role of pacing' and "Acquired third-degree (complete) atrioventricular block", section on 'Place permanent pacemaker if AV block is not reversible' and "Permanent cardiac pacing: Overview", section on 'Acquired AV block'.)

  - Sinus node dysfunction – The need for permanent pacing in patients with sinus node dysfunction depends upon the correlation of bradycardia with symptoms. There are limited pediatric data to guide management decisions, and practice is largely extrapolated from the experience in adult patients. (See "Sinus node dysfunction: Treatment", section on 'Long-term management'.)

- Congenital heart disease – Indications for permanent pacing in patients with CHD are generally similar to those in acquired heart disease. However, this group is more susceptible to having symptomatic bradycardia as they may be less able to tolerate a low heart rate due to underlying compromised cardiac function. (See 'Congenital heart disease' above and "Atrial arrhythmias and atrioventricular block associated with congenital heart disease", section on 'Permanent pacemaker implantation'.)

- Asymptomatic sinus bradycardia in otherwise healthy children – Children with sinus bradycardia who are asymptomatic and otherwise healthy generally do not require intervention. (See 'Asymptomatic bradycardia in otherwise healthy children' above.)

Indications for permanent pacemaker placement are discussed separately. (See "Permanent cardiac pacing: Overview".)

ACKNOWLEDGMENT

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Topic 5759 Version 39.0

GRAPHICS

Table 1: **Pediatric respiratory rate and heart rate lower limit, normal range, and upper limit by age***

Age	Respiratory rate (breaths/minute)			Heart rate (beats/minute)		
	Lower limit (1st percentile)	Normal range (10th to 90th percentile)	Upper limit (99th percentile)	Lower limit (1st percentile)	Normal range (10th to 90th percentile)	Upper limit (99th percentile)
0 to 3 months	25	34 to 57	66	107	123 to 164	181
3 to <6 months	24	33 to 55	64	104	120 to 159	175
6 to <9 months	23	31 to 52	61	98	114 to 152	168
9 to <12 months	22	30 to 50	58	93	109 to 145	161
12 to <18 months	21	28 to 46	53	88	103 to 140	156
18 to <24 months	19	25 to 40	46	82	98 to 135	149
2 to <3 years	18	22 to 34	38	76	92 to 128	142
3 to <4 years	17	21 to 29	33	70	86 to 123	136
4 to <6 years	17	20 to 27	29	65	81 to 117	131
6 to <8 years	16	18 to 24	27	59	74 to 111	123
8 to <12 years	14	16 to 22	25	52	67 to 103	115
12 to <15 years	12	15 to 21	23	47	62 to 96	108
15 to 18 years	11	13 to 19	22	43	58 to 92	104

* The respiratory and heart rates provided are based upon measurements in awake, healthy infants and children at rest. Many clinical findings besides the actual vital sign measurement must be taken into account when determining whether a specific vital sign is normal in an individual patient. Values

for heart rate or respiratory rate that fall within normal limits for age may still represent abnormal findings that are caused by underlying disease in a particular infant or child.

Data from: Fleming S, Thompson M, Stevens R, et al. Normal ranges of heart rate and respiratory rate in children from birth to 18 years of age: A systematic review of observational studies. Lancet 2011; 377:1011.

Graphic 78097 Version 11.0

Table 2: **Causes of bradycardia in children**

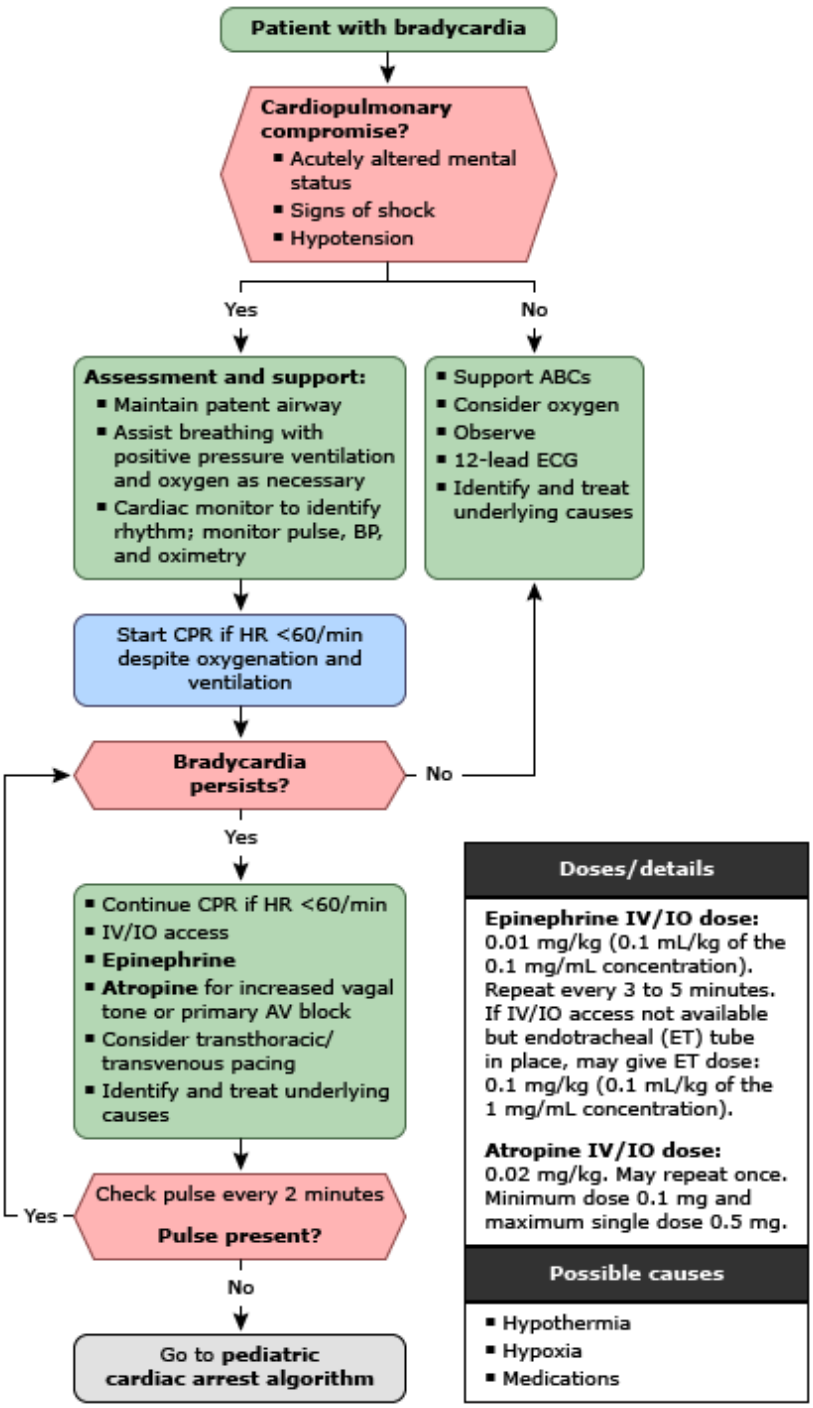
Intrinsic causes	Associated clinical and laboratory findings
Structural congenital heart disease	
Atrial septal defect	▪ Infants with a large ASD present with signs of heart failure (eg, tachypnea and dyspnea, poor feeding, failure to thrive) ▪ Physical exam findings may include fixed S_2 split and/or midsystolic pulmonary flow or ejection murmur ▪ ECG may show incomplete right bundle branch pattern
Atrioventricular canal	▪ Typically presents in early infancy with signs of heart failure (eg, tachypnea and dyspnea, poor feeding, failure to thrive) ▪ Physical exam findings may include hyperactive precordium with inferior and laterally displaced precordial impulse, increased P_2, and systolic ejection murmur due to increased blood flow across the pulmonary valve ▪ ECG shows leftward and superior QRS axis
Heritable arrhythmia syndromes	
Long QT syndrome	▪ ECG shows prolonged QTc for age, often with biphasic or notched T wave ▪ Suggestive family history (family member with LQTS or sudden cardiac death)
Brugada syndrome	▪ ECG shows shortened right ventricular conduction delay with abnormal ST segments in anterior precordial leads ▪ May be associated with sinus bradycardia and/or atrioventricular block in some cases
Collagen vascular disease	
Systemic lupus erythematosus	▪ Involvement of other organs, including skin, joints, kidneys, lungs, nervous system, serous membranes ▪ Elevated inflammatory markers (ESR and CRP)
Surgical trauma	
Sinus node dysfunction	▪ Surgical closure of ASD ▪ Fontan ▪ Atrial switch operation
Atrioventricular block	▪ Septal defect closures
Extrinsic causes	**Associated clinical and laboratory findings**
Medications*	▪ History of exposure (either via prescribed medication or accidental ingestion)
Beta-adrenergic blockers	▪ Also may cause hypotension and hypoglycemia

Calcium channel blockers	▪ Also causes hypotension, which may be severe in large overdoses
Clonidine	▪ Also may cause hypotension and sedation; in large overdoses may cause hypertension
Opioids	▪ Constricted pupils, CNS depression, respiratory depression
Hypothermia	▪ Bradycardia may occur with core temperature <35°C
Elevated intracranial pressure	▪ Associated with hypertension, abnormal respirations, and abnormal pupillary response
Hypervagotonia/neurally mediated (reflex) syncope	▪ Usually triggered by a precipitating event (eg, nasopharyngeal or esophageal stimulation, breath-holding spell, coughing, vomiting)
Sleep	▪ Heart rates during sleep can be as low as 60 to 80 bpm in infants and young children; 40 to 50 bpm in school-aged children; and 30 to 40 bpm in adolescents

ASD: atrial septal defect; S_2: second heart sound; ECG: electrocardiogram; P_2: pulmonic valve closure sound; QTc: corrected QT interval; LQTS: long QT syndrome; ESR: erythrocyte sedimentation rate; CRP: C-reactive protein; CNS: central nervous syndrome; bpm: beats per minute.
* Medications that are common causes of bradycardia in children are listed here. For a more complete list, refer to table on medications that can cause bradycardia in children.

Graphic 65427 Version 6.0

Algorithm 1: **Pediatric bradycardia with a pulse 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Bradycardia with a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ABCs: airway, breathing, circulation; AV: atrioventricular; BP: blood pressure; CPR: cardiopulmonary resuscitation; ECG: electrocardiogram; ET: endotracheal; HR: heart rate; IO: intraosseous; IV: intravenous.

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Graphic 129941 Version 11.0

Table 3: **Medications that can cause bradycardia in children**

Acetylcholinesterase inhibitors (indirect-acting parasympathomimetic)
Donepezil
Echothiophate*
Edrophonium
Galantamine
Neostigmine
Physostigmine
Pyridostigmine
Rivastigmine
Analgesic, anesthetic, sedative-hypnotic
Barbiturates (eg, phenobarbital, pentobarbital)
Benzodiazepines (eg, lorazepam, diazepam)¶
Dexmedetomidine
Local anesthetics (eg, lidocaine, mepivacaine, prilocaine)
Opioids (morphine > fentanyl)
Propofol
Antimigraine
Ergots (eg, ergotamine)
Cholinergic (direct-acting parasympathomimetic)
Acetylcholine*
Carbachol*
Cevimeline*
Cardiovascular
Alpha$_2$-adrenergic agonists (eg, clonidine, guanfacine, methyldopa)
Antiarrhythmics (eg, amiodarone, adenosine, lidocaine)
Beta-blockers
Calcium channel blockers (dihydropyridine > nondihydropyridine)
Digoxin
Nitrates (eg, nitroglycerin, nitroprusside)
Corticosteroids
Glucocorticoids (eg, methylprednisolone, prednisone)
Mineralocorticoids (eg, fludrocortisone)

Psychotropic
Lithium
Vasoactive (including topical decongestants and vasoconstrictors)*
Alpha-adrenergic agonists (phenylephrine*, midodrine, naphazoline*, oxymetazoline*, tetrahydrozoline*)
Octreotide

Medications associated with bradycardia in children. Effect may be related to dose, method, and rate of administration or exposure.

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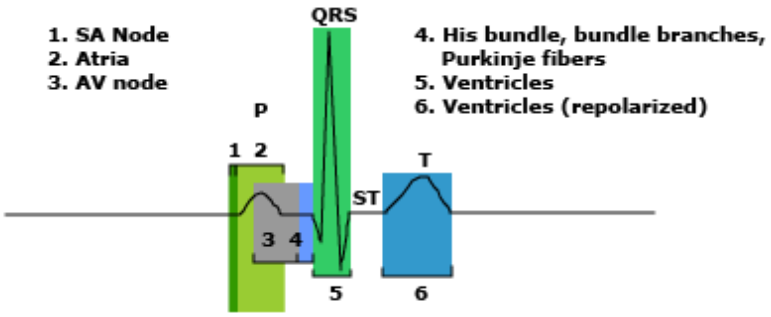
* Includes decongestant nasal sprays (imidazoline derivatives), eye drops (miotics, vasoconstrictors), and other topical vasoactive agents that may cause severe bradycardia upon oral ingestion of relatively small amounts.
¶ Bradycardia may be seen with rapid intravenous administration or overdose, or in combined use with opioid analgesics.

Graphic 102555 Version 2.0

Figure 1: **Generation of the normal electrocardiogram**



Relation between cardiac depolarization, ventricular repolarization, and the appearance of the normal electrocardiogram, including the P wave, QRS complex, ST segment, and T wave. The PR interval can be prolonged by disease from the atria to the Purkinje fibers (steps 2 to 4). The SA node is not seen on the surface ECG due to its small mass; its activity is inferred from the P wave, which reflects atrial activation.

SA: sinoatrial; AV: atrioventricular; ECG: electrocardiogram.

Data from Arnsdorf MA in: Electrophysiology of the Heart. Electrocardiography II: Applied Theory, Part 1, American Physiological Society, Bethesda 1978.

Graphic 76390 Version 5.0

Waveform 1: **Single lead electrocardiogram (ECG) showing sinus bradycardia**

Marked sinus bradycardia at a rate of 25 to 30 beats/min. The normal P waves (upright in lead II) and PR interval are consistent with a sinus mechanism with normal atrioventricular (AV) conduction.

Courtesy of Ary Goldberger, MD.

Graphic 52675 Version 4.0

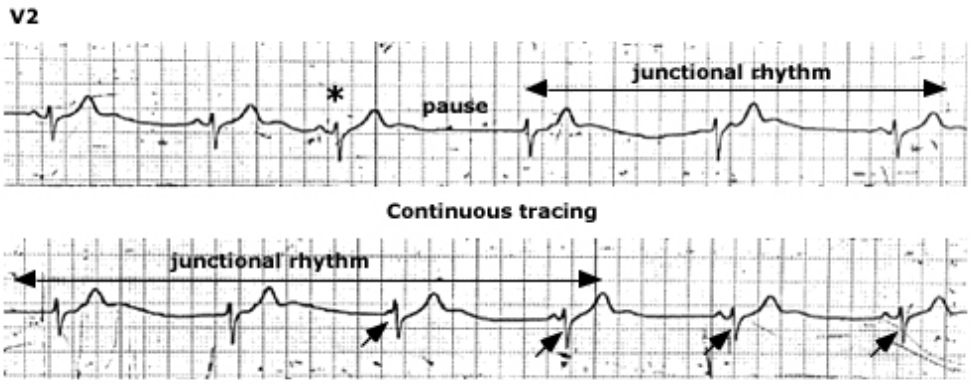
Normal rhythm strip

Normal rhythm strip in lead II. The PR interval is 0.15 sec and the QRS duration is 0.08 sec. Both the P and T waves are upright.

Courtesy of Morton F Arnsdorf, MD.

Graphic 59022 Version 3.0

Waveform 2: **ECG of sinus node dysfunction**



The continuous lead V2 rhythm strip shows evidence of sinus node dysfunction. Following the third sinus beat (*) there is a pause followed by the occurrence of an escape junctional rhythm at a slower rate. P waves reappear (arrows), which are initially dissociated from the QRS complex, but ultimately they become associated with the QRS, resulting in 1:1 conduction.

Reproduced with permission by Samuel Levy, MD.

Graphic 78649 Version 3.0

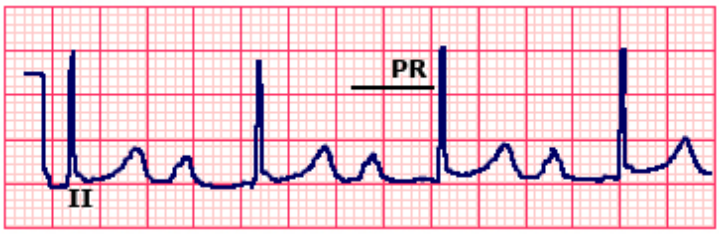
Table 4: **Major causes of atrioventricular (AV) block**

Physiologic and pathophysiologic
Increased vagal tone
Progressive cardiac conduction system disease
With fibrosis and/or sclerosis (Lenegre disease)
With calcification (Lev disease)
Ischemic heart disease, including acute myocardial infarction
Cardiomyopathy
Infiltrative processes (eg, sarcoidosis, amyloidosis, hemochromatosis, malignancy, etc)
Other non-ischemic cardiomyopathies (eg, idiopathic, infectious, etc)
Infections (eg, viral myocarditis, Lyme carditis)
Congenital AV block
Related to structural congenital heart disease
As part of neonatal lupus syndrome
Other
Hyperkalemia, severe hypo- or hyperthyroidism, trauma, degenerative neuromuscular diseases
Iatrogenic
Drugs
Beta blockers, calcium channel blockers, digoxin, adenosine, antiarrhythmic drugs
Cardiac surgery
Post valvular surgery, post surgical correction of congenital heart disease
Transcatheter aortic valve implantation
Catheter ablation of arrhythmias
Transcatheter closure of VSD
Alcohol septal ablation for HCM

VSD: ventricular septal defect; HCM: hypertrophic cardiomyopathy.

Graphic 62885 Version 6.0

Waveform 3: **Single-lead electrocardiogram (ECG) showing first-degree atrioventricular (AV) block I**



Electrocardiogram of lead II showing normal sinus rhythm, first-degree atrioventricular block with a prolonged PR interval of 0.30 seconds, and a QRS complex of normal duration. The tall P waves and P wave duration of approximately 0.12 seconds suggest concurrent right atrial enlargement.

Courtesy of Morton Arnsdorf, MD.

Graphic 67882 Version 6.0

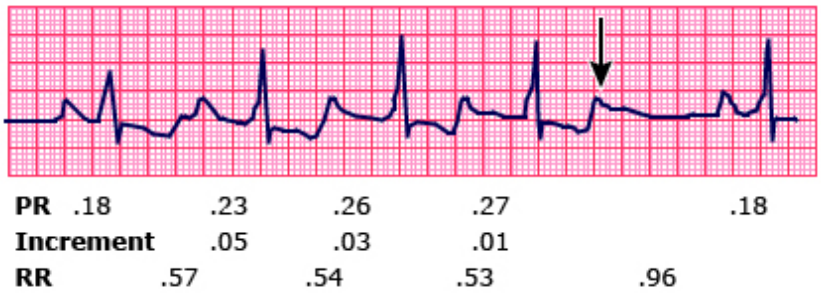
Normal rhythm strip

Normal rhythm strip in lead II. The PR interval is 0.15 sec and the QRS duration is 0.08 sec. Both the P and T waves are upright.

Courtesy of Morton F Arnsdorf, MD.

Graphic 59022 Version 3.0

Waveform 4: **Electrocardiogram showing Mobitz type I (Wenckebach) atrioventricular block**



Single-lead electrocardiogram showing Mobitz type I (Wenckebach) second-degree atrioventricular block with 5:4 conduction. The characteristics of this arrhythmia include: a progressively increasing PR interval until a P wave is not conducted (arrow), a progressive decrease in the increment in the PR interval, a progressive decrease in the RR interval, and the RR interval that includes the dropped beat (0.96 s) is less than twice the RR interval between conducted beats (0.53 to 0.57 s).

Courtesy of Morton Arnsdorf, MD.

Graphic 73051 Version 7.0

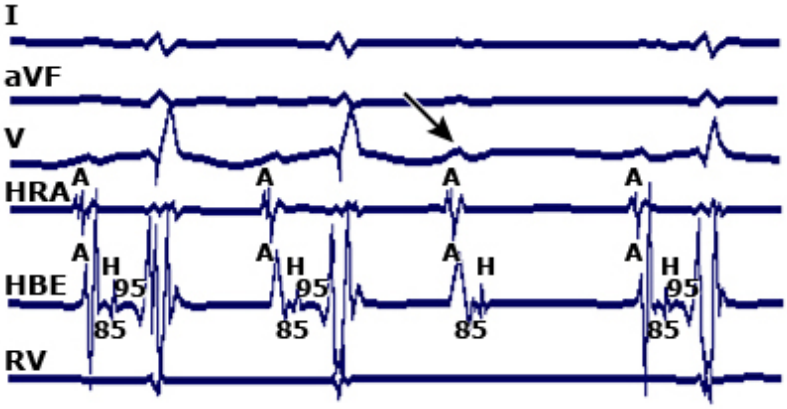
Normal rhythm strip

Normal rhythm strip in lead II. The PR interval is 0.15 sec and the QRS duration is 0.08 sec. Both the P and T waves are upright.

Courtesy of Morton F Arnsdorf, MD.

Graphic 59022 Version 3.0

Waveform 5: **Electrocardiographic and electrophysiologic features of Mobitz type II second-degree atrioventricular block**



The PR and RR intervals are constant, but the third atrial beat (A) is not conducted (arrow). His bundle electrocardiography (HBE) shows constant AH (85 ms) and HV (95 ms) intervals and normal AH but no HV conduction in the nonconducted beat. The last finding indicates that the block is distal to the His bundle, in contrast with the more proximal location of Mobitz type I atrioventricular block.

Adapted from: Josephson ME, Clinical Cardiac Electrophysiology: Techniques and Interpretations, 2nd ed, Lea & Febiger, Philadelphia 1993.

Graphic 79539 Version 8.0

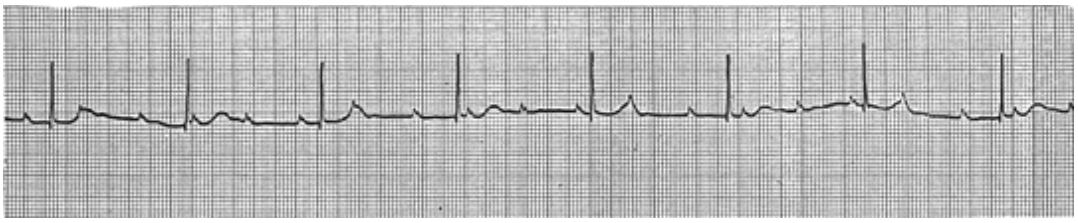
Normal rhythm strip

Normal rhythm strip in lead II. The PR interval is 0.15 sec and the QRS duration is 0.08 sec. Both the P and T waves are upright.

Courtesy of Morton F Arnsdorf, MD.

Graphic 59022 Version 3.0

Waveform 6: **Third degree atrioventricular block**



Graphic 63951 Version 3.0

Contributor Disclosures

Jennifer N Silva, MD Equity Ownership/Stock Options: Excera [Medical device (ultrasound)]; Sentiar [Medical device (augmented reality)]. Patent Holder: Excera [Medical device (ultrasound)]; Sentiar [Medical device (augmented reality)]. Grant/Research/Clinical Trial Support: Abbott [Devices]; AliveCor [Devices]; Medtronic [Devices]; UN&UP [Medical device; robotics]. Consultant/Advisory Boards: Abbott [Medical device]; AltaThera [Pharma]; Biosense Webster [Medial device]; Janssen [Pharma]. All of the relevant financial relationships listed have been mitigated. **John K Triedman, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Carrie Armsby, MD, MPH** No relevant financial relationship(s) with ineligible companies to disclose.

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